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| Inventor(s) | Lee; Roland Wing Fai et al. |

## BARCODE READER ASSEMBLY

### Abstract

An example mobile workstation that is movable by an operator includes a display device, a mounting bracket, and a barcode reader assembly. The barcode reader assembly includes a handheld barcode reader and a cradle configured to be secured to the mounting bracket. The cradle has a mechanical latch mechanism configured to removably retain the handheld barcode reader in the cradle. A first force is required to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle and a second force, greater than the first force, is required to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.

**Inventors:** **Lee; Roland Wing Fai (Jericho, NY), Srinivasan; Raghuraj (Holtsville, NY)**

**Applicant:** **ZEBRA TECHNOLOGIES CORPORATION** (Linconshire, IL)

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### Background/Summary

## BACKGROUND

[0001] Some handheld barcode reader assemblies can be used on mobile workstations. These barcode reader assemblies require that the handheld barcode reader be able to be inserted into the cradle with minimal force, but have a higher retention/removal force to retain the handheld barcode reader in the cradle as the mobile workstation is being moved and still allow the user to remove the handheld barcode reader from the cradle.

## SUMMARY

[0002] In an embodiment, the present invention is a mobile workstation that is movable by an operator. The mobile workstation comprises a display device, a mounting bracket, and a barcode reader assembly mounted to the mounting bracket. The barcode reader assembly includes a handheld barcode reader and a cradle configured to be secured to the mounting bracket. The cradle comprises a mechanical latch mechanism that is configured to: removably retain the handheld barcode reader in the cradle; require a first force to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle; and require a second force, greater than the first force, to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.

[0003] In a variation of this embodiment, the mounting bracket is secured to the display device and at least a portion of the mounting bracket extends above the display device.

[0004] In another variation of this embodiment, the first force is equal to or less than a weight of the handheld barcode reader.

[0005] In another variation of this embodiment, the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.

[0006] In another variation of this embodiment, the latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.

[0007] In another variation of this embodiment, the mechanical latch mechanism comprises a compression spring coupled to the latch bolt. The compression spring is configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and the latch bolt translated along the second axis.

[0008] In another variation of this embodiment, the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis. A housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.

[0009] In another variation of this embodiment, the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.

[0010] In another variation of this embodiment, the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.

[0011] In another variation of this embodiment, the mounting bracket comprises an aperture, the cradle comprises a threaded aperture, and the aperture in the mounting bracket and the threaded aperture in the cradle are configured to receive a threaded member to secure the cradle to the mounting bracket.

[0012] In another embodiment, the present invention is a barcode reader assembly configured to be

mounted on a mobile workstation that is movable by an operator. The barcode reader assembly comprises a handheld barcode reader and a cradle comprising a mechanical latch mechanism that is configured to removably retain the handheld barcode reader in the cradle. A first force is required to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle and a second force, greater than the first force, is required to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.
[0013] In a variation of this embodiment, the first force is equal to or less than a weight of the handheld barcode reader.
[0014] In another variation of this embodiment, the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.
[0015] In another variation of this embodiment, the latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.
[0016] In another variation of this embodiment, the latch mechanism comprises a compression spring coupled to the latch bolt. The compression spring is configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and the latch bolt translated along the second axis.
[0017] In another variation of this embodiment, the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis. A housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.
[0018] In another variation of this embodiment, the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.
[0019] In another variation of this embodiment, the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.
[0020] In another embodiment, the present invention is a cradle for a barcode reader assembly configured to be mounted on a mobile workstation that is movable by an operator. The cradle comprises a housing forming a cavity configured to receive a handheld barcode reader and a mechanical latch mechanism disposed within the housing and configured to removably retain the handheld barcode reader in the cradle. A first force is required to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle and a second force, greater than the first force, is required to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.
[0021] In a variation of this embodiment, the first force is equal to or less than a weight of the handheld barcode reader.
[0022] In another variation of this embodiment, the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.
[0023] In another variation of this embodiment, the latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.

[0024] In another variation of this embodiment, the latch mechanism comprises a compression spring coupled to the latch bolt. The compression spring is configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and the latch bolt translated along the second axis.
[0025] In another variation of this embodiment, the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis. A housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.
[0026] In another variation of this embodiment, the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.
[0027] In another variation of this embodiment, the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.
[0028] In another variation of this embodiment, the housing comprises a threaded aperture configured to receive a threaded member to secure the cradle to a mounting bracket.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0029] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.
[0030] FIG. **1** illustrates a side view of an example mobile workstation;
[0031] FIG. **2** illustrates a front perspective view of an example barcode reader assembly of the mobile workstation of FIG. **1**;
[0032] FIG. **3** illustrates a front perspective view of an example cradle of the barcode reader assembly of FIG. **2**;
[0033] FIG. **4** illustrates a cross-sectional view of the cradle of FIG. **3**, taken along line **4-4** of FIG. **3**;
[0034] FIG. **5** illustrates a cross-sectional view of the cradle of FIG. **3** taken along line **5-5** of FIG. **3**;
[0035] FIG. **6** illustrates a cross-sectional view of the cradle of FIG. **3** taken along line **6-6** of FIG. **3**;
[0036] FIG. **7** illustrates a top view of an example mechanical latch mechanism of the cradle of FIG. **3**;
[0037] FIG. **8** illustrates a side view of the mechanical latch mechanism of FIG. **7**;
[0038] FIG. **9** illustrates a cross-sectional view of the cradle of FIG. **3**, taken along line **4-4** of FIG. **3**, with an example handheld barcode reader being inserted into the cradle;
[0039] FIG. **10** illustrates a partial cross-sectional view of the cradle of FIG. **3**, taken along line **5-5** of FIG. **3**, with the example handheld barcode reader being inserted into the cradle;
[0040] FIG. **11** illustrates a cross-sectional view of the cradle of FIG. **3**, taken along line **4-4** of FIG. **3**, with the example handheld barcode reader fully inserted into the cradle;
[0041] FIG. **12** illustrates a partial cross-sectional view of the cradle of FIG. **3**, taken along line **5-5**

of FIG. **3**, with the example handheld barcode reader fully inserted into the cradle;

[0042] FIG. **13** illustrates a cross-sectional view of the cradle of FIG. **3**, taken along line **4-4** of FIG. **3**, with the example handheld barcode reader being removed from the cradle; and

[0043] FIG. **14** illustrates a partial cross-sectional view of the cradle of FIG. **3**, taken along line **5-5** of FIG. **3**, with the example handheld barcode reader being removed from the cradle.

[0044] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity, have not necessarily been drawn to scale, and that details that are not necessary for an understanding of the invention or that render other details difficult to perceive may be omitted. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0045] The apparatus components have been represented where appropriate by conventional symbols in the drawings, showing only those components and specific details that are pertinent to understanding the examples of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

## DETAILED DESCRIPTION

[0046] The example mobile workstations, barcode reader assemblies, and cradles disclosed herein utilize a mechanical latch mechanism in the cradle to provide a high retention/removal force for the handheld barcode reader in the cradle, while also providing an asymmetric lower insertion force to allow the handheld barcode reader to be able to be inserted into the cradle with minimal force. Therefore, when mounted on a mobile workstation, such as those mounted sideways over a monitor in a medical facility, the handheld barcode scanner can be easily inserted into the cradle by the user, but will be retained in the cradle as the mobile workstation is moved.

[0047] To provide these asymmetric forces, the mechanical latch mechanism disclosed herein includes a latch bolt that extends into a cavity of the cradle. The latch bolt is limited to rotational movement when the handheld barcode reader is inserted into the cradle, which bends a compression spring coupled to the latch bolt, requiring minimal force. Conversely, the latch bolt is limited to translational movement when the handheld barcode reader is removed from the cradle, which compresses the compression spring, requiring a greater force. The spring constant of the compression spring can be chosen to provide the desired insertion and retention/removal forces required for a particular application.

[0048] Referring to FIG. **1**, an example mobile workstation **100** is illustrated. Mobile workstation **100** is movable by an operator and generally includes a cart **102**, a computing device **120**, a display device **122**, an input device **124**, a mounting bracket **130**, and a barcode reader assembly **200**.

[0049] In the example shown, cart **102** includes a base **104** having a plurality of wheels **106**, a tower **108** extending generally vertically from base **104**, and a work surface **110** extending from tower **108**. Tower **108** can be extendable vertically to allow adjustment of the vertical position of work surface **110**. A first handle **112** can be mounted to tower **108** and can extend towards a rear of cart **102** and a second handle **114** can be mounted to work surface **110** or tower **108** and can extend towards a front of cart **102** to allow the operator to move mobile workstation **100** from the front or rear.

[0050] Computing device **120** can be positioned on work surface **110**, or otherwise mounted on work surface **110** or tower **108**, and can be a desktop computer or any type of computing device having at least a processor and memory. Display device **122** is mounted to tower **108**, can be a monitor or any other type of display, and is communicatively coupled to computing device **120** via a wired or wireless connection. Input device **124** can be positioned on work surface **110**, can be a keyboard, a mouse, a touchpad, or any other type of input device, and is also communicatively coupled to computing device **120** via a wired or wireless connection. In some implementations, display device **122** and input device **124** can be combined into a single device, for example, a touchscreen display. In other implementations, computing device **120**, display device **122**, and input device **124** could be combined into a single device, for example, a laptop or tablet.

[0051] Mounting bracket **130** is secured to display device **122** and/or tower **108** and barcode reader assembly **200** is mounted to mounting bracket **130**. Preferably, mounting bracket **130** is secured to display device **122** and/or tower **108** so that at least a portion of mounting bracket **130** extends above display device **122** and at least a head **302** of handheld barcode reader **300** extends past a front of display device **122** with barcode reader assembly **200** mounted to mounting bracket **130**. In this configuration, a field-of-view (FOV) **304** of handheld barcode reader **300** would be directed generally downward in front of display device **122**.

[0052] Referring to FIG. **2**, barcode reader assembly **200** generally includes a handheld barcode reader **300** and a cradle **400**. Handheld barcode reader **300** can generally be any handheld barcode reader that includes head **302** and a handle **306** that extends from a head **302**, such as the Zebra® DS81. In the example shown, handle **306** has a first end **308** that is adjacent head **302** and a second end **310** that is opposite first end **308** and is configured to be inserted into cradle **400**. In some implementations, handle **306** can also include a trigger **312** that can be used to activate barcode reader assembly **200**.

[0053] As shown in FIGS. **3-8**, cradle **400** of barcode reader assembly **200** generally includes a housing **402** that forms a cavity **406** that is configured to receive, and to removably retain, handheld barcode reader **300**. In the example shown, cavity **406** is formed in an upper housing portion **404** of housing **402**, which is secured to a lower housing portion **408**, for example, with threaded members. Cradle **400** is configured to be secured to mounting bracket **130**, for example, by one or more threaded members **140** that extend through an aperture(s) **132** in mounting bracket **130** and are threaded into a threaded aperture **410** in lower housing portion **408** in housing **402** of cradle **400**.

[0054] Cradle **400** includes a mechanical latch mechanism **430**, which is configured to allow the insertion and removal of, and to removably retain, handheld barcode reader **300** in cradle **400**. As described in more detail below, mechanical latch mechanism **430** is configured to require a first force to move handheld barcode reader **300** past mechanical latch mechanism **430** and insert handheld barcode reader **300** fully into cradle **400** and a second force, greater than the first force, to move handheld barcode reader **300** past mechanical latch mechanism **430** and remove handheld barcode reader **300** from cradle **400**. The asymmetry of the lower first force (insertion force) and the higher second force (removal force) allows handheld barcode reader **300** to be easily inserted into cradle **400** with minimal force, while providing sufficient force to retain handheld barcode reader **300** in cradle **400** during use and movement of barcode reader assembly **200** on mobile workstation **100**. In one implementation, the first force can be about equal to or less than a weight of handheld barcode reader **300** (e.g., the mass of handheld barcode reader 300×9.8 m/s.sup.2), which would require an operator to exert minimal force on handheld barcode reader **300** to insert handheld barcode reader **300** into cradle. Cradle **400** can also include a magnet **470** positioned within housing **402** that can be configured to attract a portion of handheld barcode reader **300** (e.g., second end **310** of handle **306**) when handheld barcode reader **300** is being inserted into and is fully inserted into cradle **400** to provide assistance to the operator and ease the insertion of handheld barcode reader **300** into cradle **400**. In addition, the second force can be greater than or equal to two times the weight of handheld barcode reader **300** (e.g., 2×the mass of handheld barcode reader 300×9.8 m/s.sup.2) and less than or equal to four times the weight of handheld barcode reader **300** (e.g., 4×the mass of handheld barcode reader 300×9.8 m/s.sup.2) to provide sufficient force to hold handheld barcode reader **300** in cradle **400** during movement of mobile workstation **100**, while still allowing the operator to remove handheld barcode reader **300** from cradle **400** without excessive force, when necessary. Other ranges for the second force are also possible, depending on a particular contemplated implementation and usage.

[0055] To retain handheld barcode reader **300** in cradle **400** and to provide the desired asymmetrical insertion and removal forces, in the example shown, mechanical latch mechanism

**430** includes a latch bolt **432** and a compression spring **460** coupled to latch bolt **432** that, as described in more detail below, is configured to bias latch bolt **432** into a first position. Referring to FIGS. **7-8**, latch bolt **432** has a main body **434** that has a first face **436** and a second face **438**, which extend into cavity **406** of housing **402**. A recess **440** is formed in main body **434** and includes a protrusion **442** that is configured to receive a first end **462** of compression spring **460**. A second end **464** of compression spring **460** engages a similar protrusion formed in lower housing portion **408** of housing **402**, to secure compression spring **460** between latch bolt **432** and housing **402**. Latch bolt **432** also includes a pair of opposing key sections **444** that extend in opposite directions from main body **434** and are configured to be received in a corresponding aperture **412** formed in lower housing portion **408** of housing **402**. Each key section **444** has a generally cylindrical protrusion **446** that extends from main body **434** along a first axis **450** and is received in a corresponding arcuate portion **414** of aperture **412** in housing **402**, which allows latch bolt **432** to rotate about first axis **450** when handle **306** of handheld barcode reader **300** is inserted into cradle **400** and the first force is exerted by handle **306** on first face **436** of latch bolt **432**, as described in more detail below. Each key section **444** also has a generally linear protrusion **448** that can extend from cylindrical protrusion **446** and/or main body **434** along a second axis **452** and is received in a corresponding linear portion **416** of aperture **412** in housing **402**, which allows latch bolt **432** to translate along second axis **452** and prevents latch bolt **432** from rotating about first axis **450** when handle **306** of handheld barcode reader **300** is removed from cradle **400** and the second force is exerted by handle **306** on second face **438** of latch bolt **432**, as described in more detail below.

[0056] When handheld barcode reader **300** is fully removed from cradle **400**, compression spring **460** biases latch bolt **432** into a first position (see, e.g., FIGS. **4-5**) such that a portion of latch bolt **431** extends into cavity **406** of cradle **400**.

[0057] As handheld barcode reader **300** is inserted into cradle **400**, a side surface **316** of a base **314** at second end **310** of handle **306** of handheld barcode reader **300** engages first face **436** of latch bolt **432** and rotates latch bolt **432** about first axis **450** from the first position to a second position (see, e.g., FIGS. **9-10**). When a first force is placed on first face **436** of latch bolt **432** by base **314** and latch bolt **432** is moved from the first position to the second position, latch bolt **432** is configured to rotate about first axis **450** and compression spring **460** is configured to bend. As latch bolt **432** rotates about first axis, cylindrical protrusion **446** of key section **444** of latch bolt **432** engages and rotates within arcuate portion **414** of aperture **412** of cradle **400**. To move base **314** of handle **306** of handheld barcode reader **300** past latch bolt **432** and fully insert handheld barcode reader **300** into cradle **400**, the first force exerted by base **314** of handle **306** of handheld barcode reader **300** on first face **436** of latch bolt **432** must be greater than the bending force exerted by compression spring **460**. Preferably, the bending force exerted by the compression spring, and, therefore, the first force that must be exerted to overcome the bending force and move base **314** of handle **306** of handheld barcode reader **300** past latch bolt **432**, is about equal to or less than a weight of handheld barcode reader **300**, which would require an operator to exert minimal force on handheld barcode reader **300** to insert handheld barcode reader **300** into cradle.

[0058] When handheld barcode reader **300** is fully inserted into cradle **400**, base **314** of handle **306** of handheld barcode reader **300** no longer exerts a force on first face **436** of latch bolt **432** and compression spring **460** biases latch bolt **432** back into the first position (see, e.g., FIGS. **11-12**).

[0059] As handheld barcode reader **300** is removed from cradle **400**, a top surface **318** of base **314** of handle **306** engages second face **438** of latch bolt **432** and translates latch bolt **432** along second axis **452** from the first position to a third position (see, e.g., FIGS. **13-14**). When a second force is placed on second face **438** of latch bolt **432** by base **314** and latch bolt **432** is moved from the first position to the third position, latch bolt **432** is configured to translate along second axis **452** and compression spring **460** is configured to compress. In the first position, a top surface **449** of linear protrusion **448** on latch bolt **432** engages a bottom surface **417** of linear portion **416** of aperture **412** in housing **402**, which prevents latch bolt **432** from rotating about first axis **450** when the second force is exerted by top surface **318** of base **314** on second face **438** of latch bolt **432**. Therefore, as the second force is exerted by top surface **318** of base **314** on second face **438** of latch bolt **432**, linear protrusion **448** of key section **444** of latch bolt **432** translates along second axis **452** within linear portion **416** of aperture **412** of cradle **400**. To move base **314** of handle **306** of handheld barcode reader **300** past latch bolt **432** and fully remove handheld barcode reader **300** from cradle **400**, the second force exerted by base **314** of handle **306** of handheld barcode reader **300** on second face **438** of latch bolt **432** must be greater than the compression force exerted by compression spring **460**. Preferably, the second force that must be exerted to overcome the compression force exerted by compression spring **460** and move base **314** of handle **306** of handheld barcode reader **300** past latch bolt **432**, is greater than or equal to two times the weight of handheld barcode reader **300** and less than or equal to four times the weight of handheld barcode reader **300**. This range for the second force provides sufficient force to hold handheld barcode reader **300** in cradle **400** during movement of mobile workstation **100**, while still allowing the operator to remove handheld barcode reader **300** from cradle **400** without excessive force, when necessary. Other ranges for the second force are also possible, depending on a particular contemplated implementation and usage. Since the bending force of a compression spring is generally less than the compression force, the rotation of latch bolt **432** during insertion of handheld barcode reader **300** into cradle **400** (causing compression spring **460** to bend) and the translation of latch bolt **432** during removal (causing compression spring **460** to compress) will provide asymmetric insertion and removal forces that will require less force to insert handheld barcode reader **300** into cradle **400** and a greater force to remove handheld barcode reader **300** from cradle.

[0060] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

[0061] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0062] Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms "comprises," "comprising," "has", "having," "includes", "including," "contains", "containing" or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by "comprises . . . a", "has . . . a", "includes . . . a", "contains . . . a" does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms "a" and "an" are defined as one or more unless explicitly stated otherwise herein. The terms "substantially", "essentially", "approximately", "about" or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0063] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

## Claims

**1**. A mobile workstation movable by an operator, the mobile workstation comprising: a display device; a mounting bracket; and a barcode reader assembly mounted to the mounting bracket, the barcode reader assembly comprising: a handheld barcode reader; and a cradle configured to be secured to the mounting bracket, the cradle comprising a mechanical latch mechanism that is configured to: removably retain the handheld barcode reader in the cradle; require a first force to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle; and require a second force, greater than the first force, to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.

**2**. The mobile workstation of claim 1, wherein the mounting bracket is secured to the display device and at least a portion of the mounting bracket extends above the display device.

**3**. The mobile workstation of claim 1, wherein the first force is equal to or less than a weight of the handheld barcode reader.

**4**. The mobile workstation of claim 1, wherein the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.

**5**. The mobile workstation of claim 1, wherein the mechanical latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.

**6**. The mobile workstation of claim 5, wherein the mechanical latch mechanism comprises a compression spring coupled to the latch bolt and configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and with the latch bolt translated along the second axis.

**7**. The mobile workstation of claim 5, wherein: the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis; and a housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.

**8**. The mobile workstation of claim 7, wherein the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.

**9**. The mobile workstation of claim 1, wherein the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.

**10**. The mobile workstation of claim 1, wherein: the mounting bracket comprises an aperture; the cradle comprises a threaded aperture; and the aperture in the mounting bracket and the threaded aperture in the cradle are configured to receive a threaded member to secure the cradle to the mounting bracket.

**11**. A barcode reader assembly configured to be mounted on a mobile workstation movable by an operator, the barcode reader assembly comprising: a handheld barcode reader; and a cradle comprising a mechanical latch mechanism that is configured to: removably retain the handheld barcode reader in the cradle; require a first force to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle; and require a second force, greater than the first force, to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.

**12**. The barcode reader assembly of claim 11, wherein the first force is equal to or less than a weight of the handheld barcode reader.

**13**. The barcode reader assembly of claim 11, wherein the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.

**14**. The barcode reader assembly of claim 11, wherein the mechanical latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.

**15**. The barcode reader assembly of claim 14, wherein the mechanical latch mechanism comprises a compression spring coupled to the latch bolt and configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and the latch bolt translated along the second axis.

**16**. The barcode reader assembly of claim 14, wherein: the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis; and a housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.

**17**. The barcode reader assembly of claim 16, wherein the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.

**18**. The barcode reader assembly of claim 11, wherein the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.

**19**. A cradle for a barcode reader assembly configured to be mounted on a mobile workstation movable by an operator, the cradle comprising: a housing forming a cavity configured to receive a handheld barcode reader; a mechanical latch mechanism disposed within the housing, the mechanical latch mechanism configured to: removably retain the handheld barcode reader in the cradle; require a first force to move the handheld barcode reader past the mechanical latch mechanism and insert the handheld barcode reader fully into the cradle; and require a second force, greater than the first force, to move the handheld barcode reader past the mechanical latch mechanism and remove the handheld barcode reader from the cradle.

**20**. The cradle of claim 19, wherein the first force is equal to or less than a weight of the handheld barcode reader.

**21**. The cradle of claim 19, wherein the second force is greater than or equal to two times a weight of the handheld barcode reader and less than or equal to four times the weight of the handheld barcode reader.

**22**. The cradle of claim 19, wherein the mechanical latch mechanism comprises a latch bolt that is configured to rotate about a first axis with a first force placed on a first face of the latch bolt and to translate along a second axis, perpendicular to the first axis, with a second force placed on a second face of the latch bolt.

**23**. The cradle of claim 22, wherein the mechanical latch mechanism comprises a compression spring coupled to the latch bolt and configured to bend with the first force placed on the first face of the latch bolt and the latch bolt rotated about the first axis and to compress with the second force placed on the second face of the latch bolt and the latch bolt translated along the second axis.

**24**. The cradle of claim 22, wherein: the latch bolt comprises a key section having a generally cylindrical protrusion extending along the first axis and a linear protrusion extending from the cylindrical protrusion along the second axis; and a housing of the cradle comprises an aperture having an arcuate portion configured to receive the cylindrical protrusion of the key section of the latch bolt and a linear portion configured to receive the linear protrusion of the latch bolt.

**25**. The cradle of claim 24, wherein the aperture in the housing of the cradle allows rotation of the latch bolt about the first axis with the first force placed on the first face of the latch bolt, prevents rotation of the latch bolt about the first axis with the second force placed on the second face of the latch bolt, and allows translation of the latch bolt along the second axis with the second force placed on the second face of the latch bolt.

**26**. The cradle of claim 19, wherein the cradle comprises a magnet configured to attract a portion of the handheld barcode reader with the handheld barcode reader inserted into the cradle.

**27**. The cradle of claim 19, wherein the housing comprises a threaded aperture configured to receive a threaded member to secure the cradle to a mounting bracket.